2. (a) Stop heading and stir the water well. 1A 1A $\underline{2}$

(b) (i)

$$\frac{L-64}{80-64}=\frac{65-20}{92-20}$$ 1M

$$L-64=10$$

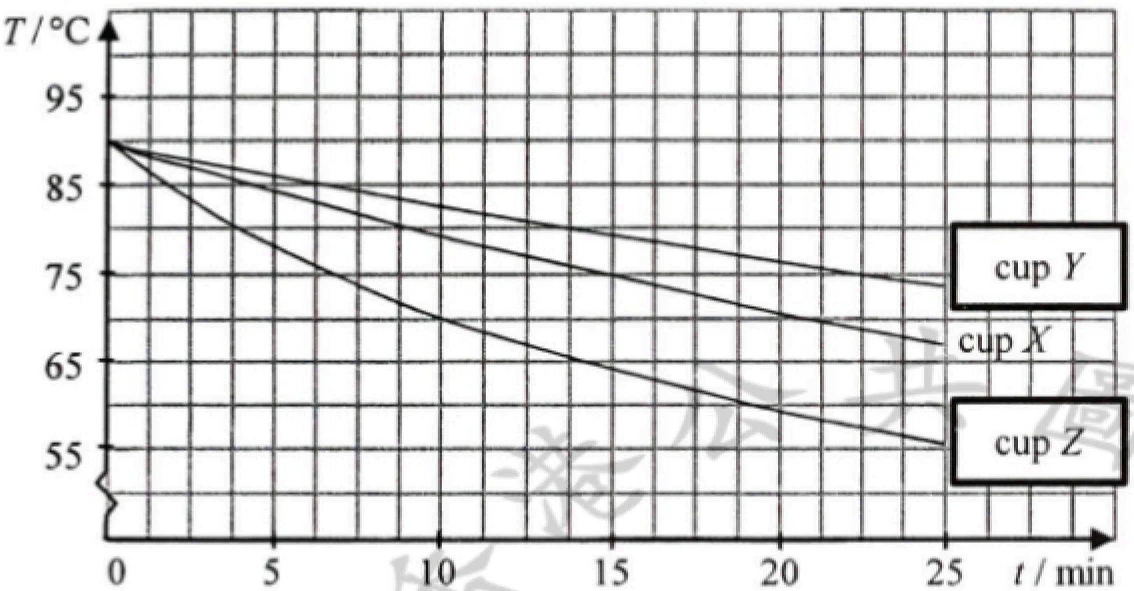
$$L=74 \text{ mm}$$ 1A $\underline{2}$

(ii) Let x °C be the absolute zero.

$$\frac{20-x}{92-20}=\frac{64-0}{80-64}$$ 1M

$$20-x=288$$

$$x=-268 \text{ °C}$$ 1A $\underline{2}$

1. (a) Fair test (otherwise the initial temperature would affect the result).	1A	
	1	
(b) Temperature difference between hot water and the surroundings (room temperature) decreases as time elapses. Rate of energy loss / heat loss becomes smaller, hence rate of temperature drop is smaller and the curve becomes less steep.	1A 1A	
	2	
(c) (i) The graphs of T against t 	1A	
	1	
(ii) Comparing to cup X as a control, Cup Y: The shiny surface of aluminium foil is a poor emitter / good reflector of radiation (thus reduces heat lost to surroundings). Cup Z: No lid allows convection through air / surroundings. Or No lid enhances heat lost by evaporation / convection / contacting with air / surroundings.	1A 1A 1A 1A	
	3	
(d) Polystyrene / foam plastic or foam / wood	1A	
	1	